

Page Shell & Runtime Responsibilities — Migration Design

AEM 6.4 to AEM as a Cloud Service (Edge Delivery
Services) Migration

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Adobe Professional Services – Edge Delivery Services



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Page Shell & Runtime Responsibilities — Target State Design

1. Executive Summary

This section defines the explicit responsibility mapping for the page shell migration from AEM 6.4 to Edge Delivery Services. The current page shell owns significantly more than header/footer rendering — it orchestrates the Adobe Data Layer, analytics pipeline, consent management, accessibility runtime, personalization hooks, chat integration, and eCommerce context. Each responsibility is inventoried, its target-state owner assigned, and parity validation defined.

2. Source State: Page Shell Responsibility Inventory

2.1 Runtime Components Identified

#	Responsibility	Source Technology	Load Phase	Critical
1	Adobe Data Layer (digitalData)	Custom JS object on window	Synchronous (head)	Yes
2	Adobe Launch (Tags)	_satellite via assets.adobedtm.com	Async (head)	Yes
3	Adobe Alloy (Web SDK)	alloy namespace	Async (Launch rule)	Yes
4	TrustArc/TrustE Consent Manager	consent.truste.com script	Sync (head, blocking)	Yes
5	UserWay Accessibility Widget	cdn.userway.org widget	Async (head)	Yes
6	Skip Links & ARIA Landmarks	UserWay dynamic injection	Post-load	Yes
7	Kampyle Feedback Widget	KAMPYLE_ONSITE_SDK	Delayed	No
8	Chat (TF Chat Launcher)	chat-api.thermofisher.com	Delayed	No
9	Cart Context / eCommerce	DM Preload + REST API	Async (early)	Yes
10	Merchandising Tracking (cid)	digitalData.merchandising	Sync	Yes
11	Search Bar (v2)	Custom searchbar.js + Endeca	Async	No
12	Promo/Offer Banner (Header Bar)	Dynamic content injection	Async	No

2.2 Adobe Data Layer Structure (**window.digitalData**)

The current **digitalData** object is the single source of truth for all analytics, personalization, and tracking. It is populated synchronously before Adobe Launch fires.

```
window.digitalData = {
  page: {
    country: "us", // Geo context
    language: "en", // Language
    siteName: "thermo", // Site identifier
    type: "cq", // Page type (cq = AEM-authored)
    cqtemplate: "/apps/lifetech/templates/...", // AEM template
    jcrTitle: "Polymerase Chain Reaction (PCR)", // Page title
    ltMetaDescription: "...", // Meta description
    businessUnit: "Molecular Biology", // BU taxonomy
    searchDimensions: "SearchDimensions:PCR", // Search facets
    productDivision: "BID" // Division
  },
  user: {
    gigyaId: "", // Gigya/SAP CDC ID
    loginStatus: "anonymous" // Auth state
  },
  merchandising: {
    cid: "..." // Campaign ID tracking
  },
  cartItems: [], // Current cart state
  widgets: {
    chat: {...}, // Chat config
    userway: {...} // Accessibility config
  },
  // Methods
  getQueryParam: fn, // URL param extraction
  getValOnce: fn, // Dedup tracking
  getCookie: fn, setCookie: fn, // Cookie utilities
  getMasterDomain: fn, // Domain resolution
  formSubmitEvent: fn, // Form analytics
  modalVideoEvent: fn, // Video tracking
  customClickEvent: fn // Custom event dispatch
}
```

2.3 Adobe Launch Configuration

Property	Value
Property Name	Thermofisher.com - Required Only
Property ID	PRd3086b6d2b224c24be2f80bb4d1bbcd9
Turbine Version	29.0.0
Build Date	2026-03-10
Library URL	assets.adobedtm.com/7e08552ade3f/1a8047d2b483/launch-f46125d37e44.min.js
Alloy Namespace	alloy (Adobe Experience Platform Web SDK)
Extensions Loaded	4 rule components (RC*-source.min.js)

2.4 Consent Management (TrustArc)

Property	Value
Provider	TrustArc (TrustE)

Property	Value
Script	<code>consent.truste.com/notice?c=teconsent&...&domain=thermofisher.com</code>
Consent Element	<code>#teconsent</code> DOM element
Behavior	Blocks non-essential cookies until consent given
Country-Aware	Yes (country=us&language=en in script URL)
Categories	Required, Functional, Advertising (standard TrustArc)

2.5 Accessibility Runtime (UserWay)

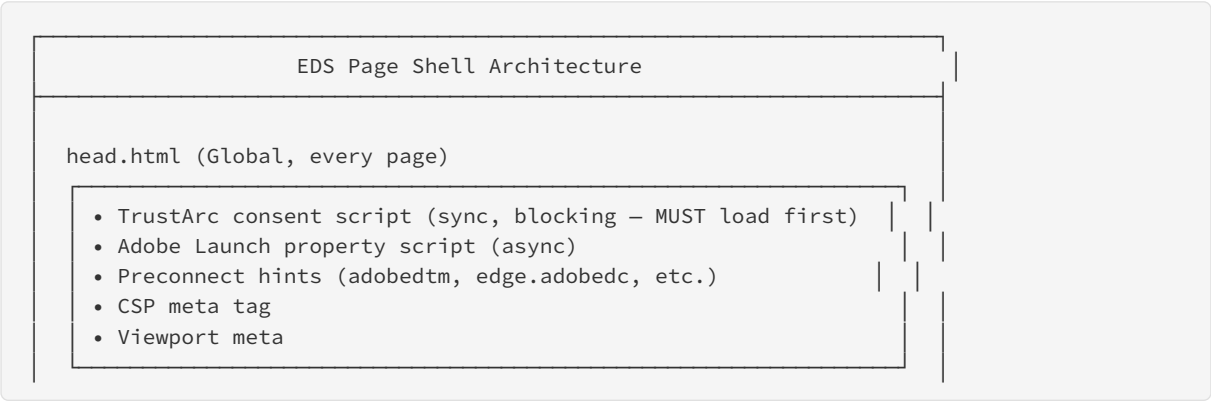
Property	Value
Provider	UserWay
Widget Script	<code>cdn.userway.org/widgetapp/.../widget_app_*.js</code>
Remediation	<code>cdn.userway.org/widgetapp/.../remediation/remediation_*.js</code>
Features	Skip-to-content, screen reader mode, accessibility menu, navigation menu
ARIA Injection	Dynamic landmark roles, skip links

2.6 Chat Integration

Property	Value
Endpoint	<code>chat-api.thermofisher.com</code>
Launcher	<code>/search/chat-launcher/chat-launcher.js</code>
App Bundle	<code>/search/chat/tf/app.*.js + /search/chat/tf/css/app.*.css</code>
Eligibility	Runtime check via API call (not all pages/users get chat)

3. Target State: EDS Responsibility Mapping

3.1 Architecture Overview





3.2 Responsibility Transfer Matrix

#	Responsibility	Source Owner	Target Owner (EDS)	Implementation	Risk if Missed
1	digitalData initialization	AEM page component (HTL)	scripts.js (eager)	Build from URL + metadata block	Analytics blackout
2	Adobe Launch loading	<head> include	head.html	Same async script tag	All tracking lost
3	Alloy/Web SDK	Launch extension	Launch extension (unchanged)	No code change — Launch handles	Data collection gap
4	TrustArc consent	<head> include	head.html	Same sync script tag (BEFORE Launch)	GDPR/CCPA violation
5	UserWay accessibility	<head> include	delayed.js	Load script dynamically	Accessibility lawsuit risk

#	Responsibility	Source Owner	Target Owner (EDS)	Implementation	Risk if Missed
6	Skip links / ARIA	UserWay auto-inject	UserWay (unchanged) + EDS semantic HTML	EDS provides native landmarks; UserWay adds overlay	WCAG failure
7	Kampyle feedback	Footer include	delayed.js	Load SDK on timer	No feedback collection
8	Chat launcher	Footer include	delayed.js	Eligibility API → inject bundle	Lost engagement channel
9	Cart preload	Page shell JS	Header block	REST API call on header load	Empty cart badge
10	Merchandising CID	digitalData init	delayed.js	Parse cid from URL params	Campaign attribution loss
11	Search bar	Header component	Header block	Searchbar v2 script integration	Broken search UX
12	Promo header bar	Dynamic component	Header block	Authored content or API-driven	Missing promotions

3.3 Load Order Dependency Chain

- | | |
|--------------------------------|---|
| 1. TrustArc consent script | ← MUST be first (blocks non-essential) |
| 2. Adobe Launch library | ← After consent, respects consent state |
| 3. digitalData initialization | ← Launch rules depend on this |
| 4. Alloy/Web SDK (via Launch) | ← Fires after digitalData ready |
| 5. Page content renders (LCP) | ← No dependency on analytics |
| 6. Header block (cart, search) | ← Lazy phase |
| 7. UserWay, Kampyle, Chat | ← Delayed phase (3s+) |

Critical invariant: TrustArc MUST load before Adobe Launch. If this order is violated, cookies may be set before consent is obtained → regulatory violation.

4. Adobe Data Layer — Target State Implementation

4.1 Data Layer Initialization Strategy

In EDS, digitalData is constructed from: 1. **URL parsing** — country, language 2. **Metadata block** — title, description, businessUnit, productDivision, template 3. **Auth state** — login status from session/cookie 4. **URL parameters** — merchandising CID, campaign tracking

4.2 Data Layer Field Mapping (Source → Target)

Field	Source (AEM 6.4)	Target (EDS)	Generation Method
page.country	Server-side from content path	URL segment parsing (/us/en/... → us)	scripts.js
page.language	Server-side from content path	URL segment parsing → en	scripts.js
page.siteName	Hardcoded in HTL	Hardcoded "thermo"	scripts.js
page.type	"cq" (AEM template type)	"eds"	scripts.js

Field	Source (AEM 6.4)	Target (EDS)	Generation Method
page.cqtemplate	JCR template path	Metadata block → Template name	scripts.js
page.jcrTitle	jcr:title property	document.title (before suffix)	scripts.js
page.ltMetaDescription	jcr:description	meta[name="description"] content	scripts.js
page.businessUnit	Page properties	Metadata block → business-unit	scripts.js
page.searchDimensions	Page properties	Metadata block → search-dimensions	scripts.js
page.productDivision	Page properties	Metadata block → product-division	scripts.js
user.gigyaId	Server-side session	Cookie/session read	scripts.js
user.loginStatus	Server-side session	Cookie presence check	scripts.js
merchandising.cid	URL param extraction	URLSearchParams → cid	delayed.js
cartItems	REST API preload	REST API call in header block	Header block

4.3 Custom Event Parity

Event Method	Purpose	EDS Implementation
digitalData.formSubmitEvent(formName, formData)	Track form submissions	Global function; dispatches CustomEvent + calls _satellite.track
digitalData.modalVideoEvent(videoId, action)	Track video play/pause/complete	Global function; same pattern
digitalData.customClickEvent(category, action, label)	Track CTA clicks, downloads	Global function; same pattern

4.4 Adobe Launch Integration Point

Adobe Launch rules are configured to fire on: - digitalData:ready custom event (page view) - digitalData.formSubmitEvent invocations - digitalData.customClickEvent invocations - DOM events (link clicks with [data-analytics] attributes)

No Launch property changes required — the Launch library remains the same. Only the data layer initialization moves from AEM HTL to EDS scripts.js.

5. Consent Management — Target State Design

5.1 TrustArc Integration in EDS

```
<!-- head.html: TrustArc MUST be the FIRST external script -->
<script src="https://consent.truste.com/notice?c=teconsent&pcookie=true&js=nj&gtm=true
  &noticeType=bb&text=true&domain=thermofisher.com&country={country}&language={lang}
  &privacypolicylink=https://www.thermofisher.com/{country}/{lang}/home/global/privacy-policy.html
```



```
&cookieLink=https://www.thermofisher.com/{country}/{lang}/home/global/how-cookies-are-used.html">
</script>
```

5.2 Consent-Aware Script Loading

Script	Consent Required	Load Behavior
TrustArc	None (required cookie)	Always loads
Adobe Launch	None (Launch itself handles consent internally)	Always loads
Alloy/Web SDK	Launch checks TrustArc state before firing beacons	Conditional via Launch
UserWay	None (accessibility = required)	Always loads
Kampyle	Functional cookies consent	Only if consent given
Chat	Functional cookies consent	Only if consent given
Merchandising tracking	Advertising cookies consent	Only if consent given

5.3 Cookie Preferences UI

The "Cookie Preferences" link in the footer triggers TrustArc's modal:

```
// Footer block must wire this on the cookie preferences link
truste.eu.clickListener();
```

6. Accessibility — Target State Design

6.1 Native EDS Accessibility (Built-in)

EDS provides strong accessibility by default:

Feature	EDS Native	Additional Work
Semantic HTML5 (<header>, <main>, <footer>, <nav>)	Yes	None
Proper heading hierarchy	Yes (from authored content)	Content authoring guidelines
Skip-to-main-content link	No (UserWay provides)	UserWay integration
Keyboard navigation	Yes (native links/buttons)	Custom widget a11y
Focus management	Partial	Block-specific focus traps for modals
ARIA landmarks	Yes (semantic elements)	None
Screen reader optimization	UserWay remediation	UserWay integration

6.2 UserWay Integration in EDS

```
// delayed.js
function loadUserWay() {
  const script = document.createElement('script');
  script.src = 'https://cdn.userway.org/widget.js';
  script.dataset.account = 'THERMO_FISHER_ACCOUNT_ID';
  document.head.appendChild(script);
}
loadUserWay();
```

6.3 Accessibility Features Preserved

Feature	Source Implementation	Target Implementation
Skip to main content	UserWay button overlay	UserWay (unchanged)
Enable screen reader mode	UserWay button	UserWay (unchanged)
Open accessibility menu	UserWay widget	UserWay (unchanged)
Open accessible navigation	UserWay button	UserWay (unchanged)
Focus visible indicators	CSS + UserWay	EDS CSS + UserWay
Reduced motion	UserWay toggle	UserWay (unchanged)

7. Chat & Feedback — Target State Design

7.1 Chat Launcher

```
// delayed.js
async function loadChat() {
  const resp = await fetch('/api/chat/eligibility');
  if (resp.ok) {
    const { eligible } = await resp.json();
    if (eligible) {
      const script = document.createElement('script');
      script.src = 'https://chat-api.thermofisher.com/search/chat-launcher/chat-launcher.js';
      document.body.appendChild(script);
    }
  }
}
loadChat();
```

7.2 Kampyle Feedback

```
// delayed.js
function loadKampyle() {
  const script = document.createElement('script');
  script.src = 'https://nebula-cdn.kampyle.com/us/wu/638829/onsite/generic.js';
  document.body.appendChild(script);
}
window.setTimeout(loadKampyle, 5000);
```

8. eCommerce Context — Target State Design

8.1 Cart Preload

The header block must preload cart state for the badge count:

```
// blocks/header/header.js
async function loadCartState() {
  try {
    const resp = await fetch('/api/store/cart/details');
    if (resp.ok) {
      const data = await resp.json();
      window.digitalData.cartItems = data.products || [];
      updateCartBadge(window.digitalData.cartItems.length);
    }
  } catch { /* silent fail – badge shows 0 */ }
}
```

8.2 Merchandising CID Tracking

```
// delayed.js or scripts.js
function initMerchandising() {
  const params = new URLSearchParams(window.location.search);
  const cid = params.get('cid') || params.get('icid');
  if (cid) {
    window.digitalData.merchandising = { cid };
    document.cookie = `cid=${cid};path=/;max-age=1800;SameSite=Lax`;
  }
}
```

9. Security Considerations

9.1 Content Security Policy

```
<!-- head.html -->
<meta http-equiv="Content-Security-Policy"
  content="script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https;
          base-uri 'self'; object-src 'none';"
  move-to-http-header="true">
```

9.2 Third-Party Script Domains Allowlist

Domain	Purpose	Required
assets.adobedtm.com	Adobe Launch	Yes
edge.adobedc.net	Adobe Alloy/Web SDK	Yes
consent.truste.com	TrustArc consent	Yes
cdn.userway.org	Accessibility widget	Yes
nebula-cdn.kampyle.com	Feedback widget	No

Domain	Purpose	Required
chat-api.thermofisher.com	Chat integration	No
dm-images.thermofisher.com	Dynamic media images	Yes

9.3 Form Security (if applicable)

Concern	Source (AEM)	Target (EDS)
CSRF tokens	AEM Forms framework	Not applicable (no server-side forms in EDS)
Form encryption	AEM Forms encryption service	Client-side validation + HTTPS only
reCAPTCHA	Component-level	Block-level integration

10. Validation & Regression Checks

10.1 Analytics Parity Test Matrix

#	Test	Expected Result	Pass Criteria	Tool
1	digitalData object present	window.digitalData defined before Launch fires	Object exists with all required keys	Browser console
2	page.country populated	Matches URL country segment	"us" for US pages	Automated check
3	page.language populated	Matches URL language segment	"en" for English	Automated check
4	page.jcrTitle populated	Matches visible H1/title	Non-empty string	Automated check
5	page.businessUnit populated	Matches metadata block value	Non-empty on authored pages	Automated check
6	_satellite loaded	Adobe Launch property active	_satellite.property.name matches	Browser console
7	Alloy sends page view	Network request to edge.adobedc.net	Beacon fired with correct data	Network tab
8	Consent blocks tracking	With no consent, Alloy must NOT fire	Zero beacons before consent	Network tab
9	Form tracking works	digitalData.formSubmitEvent dispatches correctly	Launch rule fires on form submit	Analytics debugger
10	Click tracking works	digitalData.customClickEvent dispatches correctly	Launch rule fires on CTA click	Analytics debugger
11	CID captured	URL ?cid=X populates merchandising.cid	Cookie set, data layer populated	Browser check
12	Cart state loads	Cart badge shows correct count	API returns data, badge updates	Visual + API

10.2 Accessibility Regression Tests

#	Test	Expected Result	Pass Criteria	Tool
1	UserWay widget loads	Accessibility menu button visible	Widget rendered, interactive	Visual
2	Skip-to-content works	Focus moves to main content	main receives focus on activation	Keyboard test
3	Landmark roles present	banner, main, navigation, contentinfo	All 4 present	aXe/Lighthouse
4	Keyboard navigation	All interactive elements focusable	Tab order logical, no traps	Manual test
5	Screen reader mode	UserWay SR mode activates	Content linearizes correctly	Screen reader
6	Focus indicators	Visible focus ring on all interactive elements	WCAG 2.4.7 pass	Visual test

10.3 Consent & Privacy Tests

#	Test	Expected Result	Pass Criteria	Tool
1	TrustArc loads first	Consent script precedes all other scripts	Script order in DOM	Network waterfall
2	No tracking before consent	Zero analytics beacons until accept	Network requests clean	Network tab
3	Cookie preferences modal	"Cookie Preferences" link opens TrustArc UI	Modal renders correctly	Manual test
4	Consent persists	Returning visitor not prompted again	TrustArc cookie present	Cookie inspector
5	Opt-out works	Rejecting cookies stops all non-essential tracking	Zero non-essential cookies	Cookie inspector

10.4 Performance Checks

#	Test	Expected Result	Pass Criteria
1	LCP not blocked by analytics	Analytics loads async, doesn't block render	LCP ≤ 2.5s
2	TrustArc doesn't block LCP	Consent script is small, loads fast	FCP ≤ 1.8s
3	Delayed scripts load after 3s	UserWay, Chat, Kampyle don't impact TBT	TBT ≤ 200ms
4	Total JS payload reasonable	No jQuery dependency in EDS	JS total ≤ 150KB (first-party)

11. Migration Risks & Mitigations

Risk	Impact	Likelihood	Mitigation
digitalData not ready when Launch fires	All analytics events lost (page views, clicks)	High	Initialize digitalData synchronously in scripts.js eager phase; fire digitalData:ready custom event; Launch rules use event-based trigger
TrustArc loads after Launch	Cookies set before consent = GDPR violation, potential €20M fine	Medium	Enforce script order in head.html; CSP headers; automated CI test checking script order
UserWay breaks in EDS context	Accessibility overlay fails, skip links missing	Low	UserWay is domain-agnostic; test in EDS preview before go-live
Cart API CORS issues	Cart badge shows 0, broken add-to-cart	Medium	Verify CORS headers allow EDS origins (*.aem.live, *.aem.page); add to API gateway config
Chat eligibility API blocked	Chat never loads for eligible users	Low	Verify API endpoint accepts EDS referer; add EDS domains to allowlist
CID attribution lost	Campaign ROI unmeasurable, marketing team impact	Medium	Initialize CID parsing in eager phase; persist to cookie immediately
Launch rules reference AEM-specific selectors	Rules fire on missing DOM elements → silent failure	High	Audit all Launch rules for AEM-specific selectors (.parsys, .cq-*); update to EDS equivalents

12. Implementation Checklist

#	Task	Owner	Phase	Dependency
1	Add TrustArc script to head.html (first position)	Dev	Sprint 1	None
2	Add Adobe Launch script to head.html (after TrustArc)	Dev	Sprint 1	TrustArc deployed
3	Implement digitalData initialization in scripts.js	Dev	Sprint 1	Metadata block schema finalized
4	Add business-unit, product-division, search-dimensions to metadata block schema	Content/Dev	Sprint 1	Content model defined
5	Audit Adobe Launch rules for AEM-specific selectors	Analytics team	Sprint 1	Access to Launch property

#	Task	Owner	Phase	Dependency
6	Update Launch rules to use EDS DOM selectors	Analytics team	Sprint 2	Rule audit complete
7	Implement cart preload in header block	Dev	Sprint 2	API CORS configured
8	Add UserWay to <code>delayed.js</code>	Dev	Sprint 2	Account ID confirmed
9	Add Chat launcher to <code>delayed.js</code>	Dev	Sprint 2	API endpoint CORS verified
10	Add Kampyle to <code>delayed.js</code>	Dev	Sprint 2	None
11	Add CID/merchandising tracking	Dev	Sprint 1	None
12	Wire Cookie Preferences link in footer block	Dev	Sprint 2	TrustArc deployed
13	Run full analytics parity test suite	QA	Sprint 3	All above complete
14	Run accessibility audit (aXe + manual)	QA	Sprint 3	UserWay deployed
15	Run consent flow test (GDPR, CCPA scenarios)	QA/Legal	Sprint 3	TrustArc + Launch deployed

13. Key Design Decisions

Decision	Rationale	Alternative Considered
Keep Adobe Launch unchanged	Launch rules, extensions, and property config are independent of page delivery. Moving to EDS doesn't require re-building the Launch property.	Migrate to Adobe Web SDK direct (rejected: too much re-work, Launch provides governance)
Initialize <code>digitalData</code> in eager phase	Launch rules expect the data layer before firing. Delayed initialization causes missed page views.	Initialize in <code>delayed.js</code> (rejected: creates race condition with Launch)
TrustArc in <code>head.html</code> not <code>delayed.js</code>	Legal requirement: consent must be obtained before any non-essential processing.	Load dynamically (rejected: too late, GDPR risk)
UserWay in delayed phase	Accessibility overlay is not LCP-critical; UserWay itself is ~200KB. Loading it delayed preserves performance.	Load eagerly (rejected: 500ms+ LCP regression)
EDS header block owns cart + search	These are tightly coupled to the header UI; co-locating reduces complexity.	Separate blocks (rejected: adds network waterfalls, breaks cart badge UX)
<code>page.type</code> changes from <code>"cq"</code> to <code>"eds"</code>	Allows analytics team to segment data by platform during migration.	Keep as <code>"cq"</code> (rejected: makes A/B analysis impossible)